Princeton University Mathematics Competition

PUMaC 2025 - Power Round

The Continuum Hypothesis

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Abstract: This document provides certain solutions for the PUMaC Power Round 2025. PUMaC power round consists of one of the most prominent foundations in modern mathematics, The Continuum Hypothesis, and the Axiomatic Set Theory.

Problem 1.1.1 -

$=, \in, \neg, \wedge, \exists$.

In order to show that we can rewrite the other standard logical symbols ($\vee$, $\implies$, $\iff$, $\forall$) using just the ones given above, here's my approach.

Here, we can Substitute them out one by one:

for "or" ($\vee$): we know that $A \vee B$ is equivalent to $\neg(\neg A \wedge \neg B)$.(de morgan),

for "implies" ($\implies$): we know $A \implies B$ is equivalent to $\neg A \vee B$.

since we already showed we can do $\vee$ finely with negation logic, this will work correlatively.

for "iff" ($\iff$): $A \iff B$ is just $(A \implies B) \wedge (B \implies A)$. we just use the reduction for implies.

for (for all or the / universal q.) ($\forall$): saying something is true for all x is the same as saying there doesn"t exist an x where it is already false. so $\forall x \phi$ is equivalent to $\neg \exists x(\neg \phi)$. so if we just recursivly replace all these symbols with their definitions above. And thereafter, we end up with a formula only using the allowed list.

Problem 1.2.1 -

Here, we are supposed to show that the intersection set exists.

Since X is a nonempty set, there exists some element inside it.

Let's call it $c \in X$. Which is element based, and if an element y is in the intersection of X, it must be in every element of X. since c is in X, that means y must be in as well.

So if we restrict the domain(i/o) of c using the axiom. we define the formula $\phi(y)$ to be $\forall x(x \in X \implies y \in x)$. (separation axiom)

Then by separation axiom on the set c, there exists a set $I = y \in c : \phi(y)$.

We check if this works. if $y \in I$, then by definition $\phi(y)$ is true, so y is in every element of X. conversely, if y is in every element of X, then specifically $y \in c$ because $c \in X$. and since y is in-

-every element, $\phi(y)$ is satisfied. Hence, $y \in I$. And therefore the set I satisfies the condition $y \in I \iff \forall x \in X(y \in x)$ as well.

Problem 1.2.2 -

Part A:

Applying the Pairing Axiom to the pair (x, x) this gives us the set x, x, which is equal to x and since sets do not care about repeated elements in its brackets.

Part B:

First use of pairing on x and y to get the set $A = x, y$. then we apply the Union rule of Axiom, to A. this gives us $\bigcup A$, which contains all elements that are in some member of A. since the members of are x and y, this set contains exactly the elements in x or in y.

Part C:

Using the separation axiom on x,

let the equation be $\phi(z)$ is $z \in y$. then the result of the set is $z \in x : z \in y$, which is the definition for intersection.

Problem 1.2.3 -

Forward direction is trivial:

if $x = z$ and $y = w$ just plug them in and the sets are equal. Now for the backwards part,,

Assuming that $x, x, y = z, z, w$. First look at the singletons. clearly x correlates to the relevant z because they are the smallest elements. so $x = z$, which means that, $x = z$. now looking at the other parts, we must have $x, y = z, w$.

Since we already proved $x = z$, we can take them as the same. in the equation

$x, y = x, w$, we have an x on both sides. if we ignore the x from both sets (or subtracting), we are left with y and w.

-Hence, therefore $y = w$. implies $x = z$ and $y = w$.

Problem 1.2.5 -

For this one we need to get the sets of all first coords and all second Coords.

Then, Recall that (x, y) is defined as x, x, y. since R is a set of these pairs, we can use the union axiom. $\bigcup R$ will contain all the elements of the pairs, which are sets like x and x, y.

if we apply union one more time, we get $\bigcup\bigcup R$. this set contains the elements of the elements of R, which are all the x's and y's. so this big set contains everything in the domain and range. Then f use separation on it.

$$dom(R) = x \in \bigcup\bigcup R : \exists y((x, y) \in R)$$

$$ran(R) = y \in \bigcup\bigcup R : \exists x((x, y) \in R)$$

We are filtering element of the sets from a set that we built using the Union. Final is this.

Problem 1.3.1 -

Suppose V is a Set. Then we can use the 11Separation axiom on it to make a subset. Consider $R = x \in V : x \notin x$. Since V contains all the sets, the condition $x \in V$ doesn't really restrict anything, so R is just the set of all sets that don't contain themselves.

in the result:

if $R \in R$, then $R \notin R$.

And since $R \notin R$, then so is $R \in R$.

Hence, the first assumption that V is a set that should has to be false.

Problem 1.3.2 -

First, to show $ran(F)$ is a set, This axiom basically states that if the domain of a class function is a set, then its range is also a set.

Since we are given that $dom(F)$ is a set, $ran(F)$ must also be a set. Now to show F is a set. F consists of ordered pairs (x, y) where $x \in dom(F)$ and $y \in ran(F)$.

So F is a class of the Cartesian product $dom(F) \times ran(f)$. Since both the input and output (doamain and range) are sets, their product is a set (from the earlier problem).

We can then use Separation on this product set to define F, which implies F is a set.

Problem 2.1.1 -

To show that,

$$z \in y \quad \text{and} \quad y \in x \quad \Longrightarrow \quad z \in x$$

Assume $z \in y$ and $y \in x$.

given that α is an ordinal, which means α is a transitive set.

Applying the transitivity of α to the premise $y \in x$ and $x \in \alpha$:

$$y \in x \wedge x \in \alpha \implies y \in \alpha$$

then, $z \in y$ and $y \in \alpha$. Applying the transitivity of α again:

$$z \in y \wedge y \in \alpha \implies z \in \alpha$$

Therefore, if $z \in y \in x$, then z must be an element of α.

Therefore,

Problem 2.1.3 -

Since $x, y \in \alpha$ and α is an ordinal, x and y are transitive sets, so they are ordinals as well.
Applying regularity directly to the pair $S = \{x, y\}$.

Regularity says S has an $\in$-minimal element. Which means that,
If x is minimal, $x \cap \{x, y\} = \emptyset$. This means $x \notin x$ (which we know) and $y \notin x$.
If y is minimal, $y \cap \{x, y\} = \emptyset$, so $x \notin y$ and $y \notin y$.
$x \in y$ and $y \in x$ is not possible.(both cannot be there.)

Let $I = x \cap y$.
Since x, y then,, I is also transitive.
Elements of ordinals are ordinals, so I is a set of ordinals. Thus x intersect y is an ordinal.
= $I \subseteq x$ and $I \subseteq y$.

Check if I can be a proper subset.
Suppose $I \neq x$. Then $A = x \setminus I$ is nonempty.
And A has a minimal element m. (By Regularity axiom)
$y \in x \wedge x \in \alpha \implies y \in \alpha$ $m \in x$ and $m \notin I$.
$y \in x \wedge x \in \alpha \implies y \in \alpha$ $m \cap (x \setminus I) = \emptyset$.

Any element $z \in m$ must be in x (since x is transitive).
Since $z \notin x \setminus I$, it must be that $z \in I$. ac
So $m \subseteq I$.
Since m and I belong to Ordinals, and $m \subseteq I$,,
If $m \subsetneq I$, then $m \in I$ (main property of ordinals).
If $m \in I$, then $m \in x$ and $m \in y$.

But $m \in x \setminus I$ so $m \notin y$ (actually $m \notin I$). There is a Contradiction.
So m cannot be a proper subset.
Thus $m = I$.
Since $m \in x$, $y \in x \wedge x \in \alpha \implies y \in \alpha I \in x$.
Same logic for y. If $I \neq y$, then $I \in y$.
Now, if I is a proper subset of both x and y:
Then $I \in x$ and $I \in y$.
So $I \in x \cap y$. $y \in x \wedge x \in \alpha \implies y \in \alpha$ $I \in I$.
This is a contradiction of regularity theory. So I must be equal to x or y or both.
Case x: .

Then $x \subseteq y$. Since they are distinct, $x \subsetneq y$.
Let k be minimal in $y \setminus x$.

$k \in y$ and $k \notin x$. \
$k \cap (y \setminus x) = \emptyset$.
So $k \subseteq x$.

If $k \subsetneq x$ $y \in x \wedge x \in \alpha \implies y \in \alpha$ $k \in x$. But $k \notin x$.

Here, contradiction.

So $k = x$.

Since $k \in y$, we get $x \in y$.

Case y: $I = y$.
$y \in x \wedge x \in \alpha \implies y \in \alpha$ $y \in x$.

Therefore either $x \in y$ or $y \in x$.

Problem 2.1.4 -

If $\alpha = \beta$, then $\alpha \subseteq \beta$.
If $\alpha \in \beta$, then since β is an ordinal. And so it is a transitive set. That means all its elements are also subsets of the transitive set.

So $\alpha \subseteq \beta$.
Assume $\alpha \subseteq \beta$. Now I check this: $\alpha \in \beta$ or $\alpha = \beta$.

Assume $\alpha \subsetneq \beta$,,
Let's look at the difference set $D = \beta \setminus \alpha$. Since α is a proper subset, D is not empty.

Using Axiom, pick the smallest element $m \in D$. (regularity axiom)

So $m \in \beta$ and $m \notin \alpha$.

Also, because m is minimal in D, nothing inside $m \implies$ it is inside D.
Since $m \in \beta$ and β is transitive, $m \subset \beta$.

So every element $z \in m$ is in β.

Hence.

Problem 2.2.2 -

Let ω be the set of all $\mathbb{N}$.

Assume for the sake of contradiction that C does not contain all of those elements.

$\implies$ The set of "bad" numbers $B = \{n \in \omega : n \notin C\}$ is not an empty set.

Since B is a nonempty subset of ordinals (natural numbers are also ordinals), it must have a least element. (the well ordering principle.)

Let k be the minmun element of set B (above).

$\implies k \notin C$.

Check if $k = 0$:

We are given that $0 \in C$.

$\implies k \neq 0$.

Check if k is a successor:

Since $k \in \omega$ and $k \neq 0$, k must be a successor ordinal.

$\implies \exists m \in \omega \text{ such that } k = S(m)$.

Since $m < k$ and k is the smallest element in B, m cannot be in B.

Hence, $\implies m \in C$.

But the problem premise States, if an element is in C, its successor is in C and this is another contradiction because we defined $k \in B$ (meaning $k \notin C$). So hence, our assumption is incorrect that, B set is empty.

$\implies B = \emptyset$. $\implies$ Every natural number is in C. (Proved by assuming the opposite .)

Problem 2.2.3 -

We need to form the set ω of all $\mathbb{N}$

Let $\omega = \{0, 1, 2, 3, \dots\}$.,

This set exists because of Infinity Axioms.

We need to show it is the least limit ordinal.

To confirm whether this is an Ordinal set or not, here, It contains $0, 1, 2, \cdots$, and here, each element is a transitive set (the definition of von neumann ordinals).

The set is transitive because if $n \in \omega$, then $n = \{0, 1, \dots, n-1\} \subset \omega$.

$\implies \omega$ is an ordinal.

To check whether this is a limit ordinal set and the least, we need to look for a maximum element, since any successor ordinal has a maximum.

For any $n \in \omega$, $n + 1 \in \omega$.

$n < n + 1$.

So there is no largest number.

$\implies$ not a successor ordinal. $\implies$ It is a limit ordinal.

Any limit ordinal λ must contain 0. (by definition)

If it contains n, it must contain $n + 1$ (otherwise it stops and is a successor).

So this limit ordinal must contain the entire sequence of $0, 1, 2, \cdots$. Nth element in it.

$\implies \omega \subseteq \lambda$.

$\implies \omega$ is the smallest one.

Problem 3.1.1 -

Part A:

Let $id_X : X \to X$ be $id_X(x) = x$.
id_X is injective and surjective $\implies$ bijection.
So by definition of the Definition 3.1.1. $\implies X \approx X$.

Part B:

Assume $X \approx Y$.
$\implies \exists f : X \to Y$ bijection
Inverse $f^{-1} : Y \to X$ exists and its bijection.
$\implies Y \approx X$. (from the inverse func.)

Part C:

Assume that $X \approx Y, Y \approx Z$.

$\implies \exists f : X \to Y, g : Y \to Z$ Bijections.
Define h = g of f.

Comp of 2 injections is injection; Comp of surjections is surjection.

$\implies h : X \to Z$
$\implies X \approx Z$.

Problem 3.1.3 -

Part A:

$|X| \leq |Y| \iff \exists f : X \to Y$ injection.

Assume $\exists f : X \to Y$ done injection.
Let $R = ran(f) \subseteq Y$.
Define $g : X \to R$ by $g(x) = f(x)$.
g is surjective onto R by definition.
g is injective because f is injective.
$\implies g$ is bijection $\implies X \approx R$.
Since $R \subseteq Y \implies |X| = |R| \leq |Y|$. By definition 3.1.2

($\implies$)
Assume $|X| \leq |Y|$.
Let $\kappa = |X|$ and $\lambda = |Y|$.
$\implies \kappa \subseteq \lambda$ (as ordinals).
The inclusion $\kappa \to \lambda$ is injective.
Since $X \approx \kappa$ and $Y \approx \lambda$, we can compose bijections.
Let $\phi : X \to \kappa$ and $\psi : Y \to \lambda$ be bijections.
Define $f = \psi^{-1} \circ i \circ \phi$.
Thus f is injective.
$\implies f : X \to Y$ is injective.

Part B:

$|X| \geq |Y| \iff \exists f : X \to Y$ surjection.

Assume $\exists f : X \to Y$ surjection.
Define an injection $g : Y \to X$ as follows,
For every $y \in Y$, the set $f^{-1}(\{y\}) \subseteq X$ is non-empty (since f surjective).
By Axiom of Choice, we can select one element x_y from each set $f^{-1}(\{y\})$.
Define $g(y) = x_y$.
If $g(y_1) = g(y_2) \implies x_{y_1} = x_{y_2}$.
$\implies f(x_{y_1}) = f(x_{y_2})$.

Since $x_y \in f^{-1}(\{y\}) \implies f(x_y) = y$.
$\implies y_1 = y_2$.
$\implies g$ is injective.
$\implies |Y| \leq |X|$.

($\implies$)
Assume $|X| \geq |Y|$.
$\implies |Y| \leq |X|$.
By part (a), $\exists g : Y \to X$ injection.
Define $f : X \to Y$.
For $x \in ran(g)$, let $f(x) = g^{-1}(x)$.
For $x \notin ran(g)$, let $f(x) = y_0$ (arbitrary element from Y).
$\implies f$ is surjective.

Part C:

$|X| = |Y| \iff X \approx Y$.

($\impliedby$)
Assume $X \approx Y$.
$\implies \exists f : X \to Y$ bijection.
$\implies |X| = |Y|$ by definition.

($\implies$)
Assume $|X| = |Y|$.
$\implies |X| \leq |Y|$ and $|Y| \leq |X|$.
From (a), $\exists f : X \to Y$ injection.
From (a), $\exists g : Y \to X$ injection.
Since we have injections in both directions, the sets must have the same size.
$\implies$ There exists a bijection $h : X \to Y$.

Problem 3.1.4 -

$n = 0$.

Since $0 = \emptyset$. it is an ordinal
Assume for contradictionn that 0 is not a cardinal.

$\implies \exists \alpha < 0 \text{ such that } \alpha \approx 0$.
There are no ordinals less than 0.
$\implies$ Contradiction.
$\implies |0| = 0$.
$\implies 0$ is a cardinal.

Assume n is cardinal.
Note $n + 1 = S(n) = n \cup \{n\}$.
Assume for contradiction that $n + 1$ is not a cardinal.
Hence,
$\implies |n + 1| < n + 1$.

Let $|n + 1| = k$, where k is a cardinal and $k < n + 1$.
Since k is an ordinal and $k < n + 1 = S(n) \implies k \leq n$.
Since $|n + 1| = k \implies \exists f : n + 1 \to k$ bijection.
Consider the restriction $g = f|_n$ (function restricted to domain n).
Domain of g is n. Codomain is k.

Also $k \leq n$ f.

Case 1:$k < n$.

We have an injection from a set of size n to a set of size $k < n$.
This contradicts the pigeonhole principle for finite sets.
$\implies$ Contradiction.

Case 2: $k = n$.

Then $f : n + 1 \to n$ is a bijection.
$\implies$ It is injective.
Pigeonhole Principle says we cannot map $n + 1$ items into n slots injectively.
$\implies$ Contradiction.

$\implies$ Assumption that $|n+1| < n+1$ is false.
$\implies$ $|n+1| = n+1$.
$\implies$ $n+1$ is a cardinal.

By induction, all natural numbers are cardinals.

Problem 3.2.1 -

$$\left|\bigcup_{i\in I} X_i\right| = \sum_{i\in I} |X_i|.$$

Let $\kappa_i = |X_i|$.

Definition of Sum: $\sum \kappa_i = \left|\bigsqcup_{i\in I} \kappa_i\right|$.

Definition of Disjoint Uni: $\bigsqcup X_i = \bigcup_{i\in I}(X_i \times \{i\})$.

We define a function (F) from the disjoint union to the union.

Because the (X_i) is pairwise disjoint,

$\implies \forall i, j \in I, i \neq j \implies X_i \cap X_j = \emptyset$.

Consider this union $U = \bigcup_{i\in I} X_i$.

Define $F : \bigsqcup X_i \to U$.

Let $z \in \bigsqcup X_i$. Then $z = (x, i)$ where $x \in X_i$.

Define $F((x,i)) = x$.

Assume $F((x,i)) = F((y,j))$.

$\implies x = y$.

We must show $i = j$. (indices)

Since $x \in X_i$ and $y \in X_j$, and $x = y$, we have element $x \in X_i \cap X_j$.

If $i \neq j$, then $X_i \cap X_j = \emptyset$, so x cannot exist.

$\implies i$ must equal j.

$\implies (x,i) = (y,j)$.

$\implies F$ is injective.

Let $y \in U$.

$\implies \exists k \in I \text{ such that } y \in X_k$.

Consider the element $(y,k) \in X_k \times \{k\}$.

$(y,k) \in \bigsqcup X_i$.

$F((y,k)) = y$.

$\implies F$ is Surjective.

F is a bijection.

$\implies \left|\bigsqcup X_i\right| = \left|\bigcup X_i\right|$.

Hence,

$\implies \sum |X_i| = \left|\bigcup X_i\right|$.

Problem 3.2.3 -

Let $\kappa_i = |X_i|$.

By definition of cardinality, for each i, $X_i \approx \kappa_i$.

$\implies \forall i \in I, \exists f_i : X_i \to \kappa_i$ bijection.

Let $P = \prod X_i$ (Cartesian product of sets).

Let $Q = \prod \kappa_i$ (Cartesian product of cardinals).

Define $\Phi : P \to Q$.

An element of P is a function x with $x(i) \in X_i$ for each $i \in I$.

Define $\Phi(x)$ to be a sequence y where $y(i) = f_i(x(i))$.

Assume $\Phi(a) = \Phi(b)$ for two sequences $a, b \in P$.

$\implies \forall i \in I, f_i(a(i)) = f_i(b(i))$.

Since each f_i is a bijection, it is injective.

$\implies \forall i \in I, a(i) = b(i)$.

$\implies a = b$.

$\implies \Phi$ is injective.

Let $y \in Q$. Then $\forall i, y(i) \in \kappa_i$.

Since f_i is surjective, $\exists x_i \in X_i$ such that $f_i(x_i) = y(i)$.

Construct sequence x by $x(i) = x_i$.

$\Phi(x) = y$.

$\implies$ Φ is surjective.

Φ is a bijection between P and Q.

$\implies \left|\prod X_i\right| = \left|\prod \kappa_i\right|$.

$\implies \left|\prod X_i\right| = \prod |X_i|$.

Problem 3.2.4 -

Part A: Taking Sums

Let A be a set of order type m. Let B be a set of order type n.

The ordinal Sum places all elements of B after those of A

Since A and B are finite sets, the total number of elements is just the integer sum $m + n$.

Take disjoint sets A, B with size m, n.

Take the Union.

Take the cardinality.

Since finite sets have unique order type, the sum equals $m + n$

$\implies$ The result is just the integer $m + n$ in both cases.

Part B: Products

ordinal product is:m copies of n.

order type is $n + n + \cdots + n$ (m times repeats).

cardinal product: Cartesian product $m \times n$.

The set of pairs $\{(i, j) : i < m, j < n\}$.

The number of elements in the Cartesian product is mn (integer multiplication).

The number of elements in the repeated sum is also mn.

$\Longrightarrow$ The values are equal.

References**:**

Cantor, G., 1878–1890, or Jech, T., Set Theory, 3rd edition, 2003.

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